

Ch-3

As per WHO, health is defined as an 'state of complete physical, mental & social well-being'. A healthy person does tasks efficiently & copes well in hard situations. They are also able to make better relations & adjust well with groups.

Ancient Indian medical lineage (or system), Ayurveda also describes these 3 things for better health: Dinacharya (Daily routine), Ritucharya (Seasonal routine) & Eating acc. to one's ~~own~~ prakriti (Body Constitution).

Ways for Good Health:

- Balanced diet
- Being physically active & exercising
- Adequate Sleep & limiting screen time
- Meditation
- Saying "no" to harmful substances like alcohol

AQI (Air Quality Index) is a system for measuring how clean the air is.

A symptom is what we feel (like pain) but a sign is something that can be measured (high body temp. in fever). They help doctors understand our sickness better.

Diseases

~~#~~ Non-communicable

A disease that can not through interactions with the sick person. They are linked to lifestyle and/or environment.

Eg - Diabetes, Cancer

Communicable

A disease that can spread from person to person. They are caused by pathogens.

Eg - COVID-19, Influenza

NCDs are on a rise in recent years due the lifestyle of people becoming more sedentary & food choices.

Pathogens can enter our body through air, contaminated food & water, direct contact with infected person. Some insects carry pathogens & spread them.

they are called vectors.

Common diseases by:

↳ Virus

- Common cold & Influenza → Respiratory tract
- Measles चैचकू (खसरा) → Nasal passage, skin
- Chickenpox (चैचक) → " "
- Hepatitis A → Liver
- Dengue → Blood, Skin (Transmitted by vectors)

↳ Bacterias

- TB → Lungs
- Cholera → Intestine
- Typhoid → "

↳ Protozoa

- Malaria → Skin & blood (Transmitted by ~~vectors~~ vectors)

↳ Worms

- Round worms (Ascariasis) → Intestine

★ Worms in our body are called parasites.
It means an ~~organism~~ organism living in/
living on another organism. It sucks
our nutrition & results in poor growth.

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Disease caused by the lack of a specific nutrient in the body are called Deficiency diseases. They are also NCDs.
Eg - Scurvy

Any disease that persists for a long time (more than 3 months) is called a chronic disease. Eg - Cancer

Sanitation prevents diseases by maintaining clean surroundings, proper waste disposal, safe & hygiene can reduce spread of harmful microbes & risk of outbreaks.

The natural ability of our body to resist & fight disease is called Immunity. Vaccines are made up from weakened or dead pathogens. It protects us by training our immune system to fight one or more specific diseases. Eg - BCG. Vaccines give us ~~Acq~~ Acquired Immunity. India's one of the world's largest producers of vaccines & had a key role in producing COVID-19 vaccine. They're mostly preventive

Antibiotics are the medications given when ~~some~~ someone has an bacterial disease/infection. They don't work on viral or protozoan pathogens. They are mostly curative.

Antibiotic resistance means the phenomenon in which the bacteria previously killed by Antibiotics, have built a resistance against it & won't die. To prevent this, take antibiotic only on doctor's advice & don't overuse it. It can transfer to future generations also. ~~They also~~ It also spreads via community. Eg - Farm animals develop this, so the products from their body & crops grown through their manure will transfer this resistance too, if you eat those products or crops.

The First Vaccine

- Smallpox was a dangerous disease that killed millions. There was a milder disease too, cowpox, which occurred in cows but transferred to the people, who were constantly interacting with cows.

- English Doctor, Edward Jenner, observed that people with cowpox did not get small so his hypothesis was content in the pus of cowpox blisters protected from smallpox.
- He then experimented ~~&~~ on a small boy by injecting cowpox sap in his body, who later showed no signs of smallpox. Thus, the first vaccine was discovered & mass production & usage of this, completely eradicated smallpox till 1980s.

First Antibiotic

- Pencillin (The first antibiotic) was made by Sir Alexander Fleming, who noticed that the mould on the petri dish killed the bacteria. This led the discovery of the first antibiotic. ~~&~~ (Fleming was a bacteriologist from London.)

Indian Scientists

Maharaj Kishan Bhan: He was a well-known doctor & scientist & a ~~Secret~~ Secretary of Department of Biotechnology. He wanted to make ~~heat~~ healthcare affordable & developed the Rotavirus vaccine, which protects kids from diarrhoea.

Kamal Ranadive: She studied how certain hormones & viruses are linked to cancer. Showed that tobacco, diet & pollution raises the risk of cancer, & emphasized on healthy lifestyle.